

Symmetric polynomials in free field theories

Simon Wood

The Australian National University
Department of Theoretical Physics

Joint work with Akihiro Tsuchiya ([arXiv:1302.6435](#)) and ongoing work with David Ridout

05 June 2014

Outline

- 1 Chiral algebra basics
- 2 Free field theories and screening operators
- 3 Symmetric polynomials
- 4 Application to the minimal models

Outline

- 1 Chiral algebra basics
- 2 Free field theories and screening operators
- 3 Symmetric polynomials
- 4 Application to the minimal models

Vertex operator basics

Def: VOA, data

A VOA $(V, |0\rangle, T, Y)$ is a collection of data:

- $\mathbb{N}_0$ -graded vector space $V = \bigoplus_{n \geq 0} V[n]$
- vacuum vector $|0\rangle \in V[0]$
- energy momentum vector $T \in V[2]$
- field map $Y : V \otimes V \rightarrow V((z)), a, b \mapsto Y(a; z)b,$
 $Y(a; z) = \sum_n a_n z^{-n-h_a}, a \in V[h_a]$

Vertex operator basics

Def: VOA, axioms

The VOA data satisfies the axioms:

- For $a \in V[h_a]$, $a_n : V[m] \rightarrow V[m - n]$
- $Y(|0\rangle; z) = 1$, $Y(T; z) = \sum_n L_n z^{-n-2}$
 $[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}m(m^2 - 1)\delta_{m+n,0}$
- $V[n]$ is eigenspace of L_0 with eigenvalue n .
 L_{-1} is the translation operator $[L_{-1}, Y(a; z)] = Y(L_{-1}a; z) = \partial_z Y(a; z)$
- Locality: $\exists N \in \mathbb{N}$ s.t.
 $(z - w)^N [Y(a; z), Y(b; w)] = 0$ in $\text{End}(V)[[z, z^{-1}, w, w^{-1}]]$

Vertex operator basics

Def: VOA, axioms

The VOA data satisfies the axioms:

- For $a \in V[h_a]$, $a_n : V[m] \rightarrow V[m - n]$
- $Y(|0\rangle; z) = 1$, $Y(T; z) = \sum_n L_n z^{-n-2}$
 $[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}m(m^2 - 1)\delta_{m+n,0}$
- $V[n]$ is eigenspace of L_0 with eigenvalue n .
 L_{-1} is the translation operator $[L_{-1}, Y(a; z)] = Y(L_{-1}a; z) = \partial_z Y(a; z)$
- Locality: $\exists N \in \mathbb{N}$ s.t.
 $(z - w)^N [Y(a; z), Y(b; w)] = 0$ in $\text{End}(V)[[z, z^{-1}, w, w^{-1}]]$

Axioms imply operator product expansion

$$Y(a; z)Y(b; w) = Y(Y(a; z - w)b; w) = \sum_n Y(a_n b; w)(z - w)^{-n-h_a}$$

Example: Virasoro VOA

- central charge $c \in \mathbb{C}$
- $V = U(\bigoplus_{n \geq 2} \mathbb{C}L_{-n})|0\rangle$
- vacuum $|0\rangle$, $L_{-1}|0\rangle = 0$, $T = L_{-2}|0\rangle$
- $Y(L_{-n-2}|0\rangle; z) = \frac{\partial^n}{\partial^n} Y(T; z)$, $n \geq 0$
- $Y(L_{-n_1-2} \cdots L_{-n_i-2}|0\rangle; z) =: \left(\frac{\partial^{n_1}}{\partial^{n_1}} Y(T; z) \right) \cdots \left(\frac{\partial^{n_i}}{\partial^{n_i}} Y(T; z) \right) :$

Def: (graded) VOA module

- $\mathbb{C}$ -graded vector space $M = \bigoplus_{h \in \mathbb{C}} M[h]$, $M[h]$ generalised L_0 eigenspace, eigenvalue h .
- $Y^M : V \otimes M \rightarrow M((z))$ with “obvious” generalisations of VOA axioms.

Examples

- V is a V -module.
- A proper subspace I that is a V -module is an ideal of V .
- If $|0\rangle, T \notin I$ then V/I is a VOA.

Zhu's algebra and the classification of VOA modules

Def: Zhu's algebra

- Let $O(V)$ be the span of $\text{Res}(Y(a; z)b \frac{(1+z)^{h_a}}{z^2})$, $a \in V[h_a]$, $b \in V$.
- Zhu's algebra $A(V) = V/O(V)$ is a unital associative algebra with product induced by $a * b = \text{Res}(Y(a; z)b \frac{(1+z)^{h_a}}{z})$.

Thm: Zhu

Zhu's algebra and the classification of VOA modules

Def: Zhu's algebra

- Let $O(V)$ be the span of $\text{Res}(Y(a; z)b \frac{(1+z)^{h_a}}{z^2})$, $a \in V[h_a]$, $b \in V$.
- Zhu's algebra $A(V) = V/O(V)$ is a unital associative algebra with product induced by $a * b = \text{Res}(Y(a; z)b \frac{(1+z)^{h_a}}{z})$.

Thm: Zhu

- unit $[|0\rangle] = 1$, T in centre.

Zhu's algebra and the classification of VOA modules

Def: Zhu's algebra

- Let $O(V)$ be the span of $\text{Res}(Y(a; z)b \frac{(1+z)^{h_a}}{z^2})$, $a \in V[h_a]$, $b \in V$.
- Zhu's algebra $A(V) = V/O(V)$ is a unital associative algebra with product induced by $a * b = \text{Res}(Y(a; z)b \frac{(1+z)^{h_a}}{z})$.

Thm: Zhu

- unit $[[0]] = 1$, T in centre.
- Let M be a V -module with a least conformal weight h_0 . $\overline{M} = M[h_0]$, then $[a] \in A(V)$ acts on $\overline{M}$ as a_0 and $[a * b]$ as $a_0 b_0$.

Zhu's algebra and the classification of VOA modules

Def: Zhu's algebra

- Let $O(V)$ be the span of $\text{Res}(Y(a; z)b \frac{(1+z)^{h_a}}{z^2})$, $a \in V[h_a]$, $b \in V$.
- Zhu's algebra $A(V) = V/O(V)$ is a unital associative algebra with product induced by $a * b = \text{Res}(Y(a; z)b \frac{(1+z)^{h_a}}{z})$.

Thm: Zhu

- unit $[|0\rangle] = 1$, T in centre.
- Let M be a V -module with a least conformal weight h_0 . $\overline{M} = M[h_0]$, then $[a] \in A(V)$ acts on $\overline{M}$ as a_0 and $[a * b]$ as $a_0 b_0$.
- There exists a $1 \leftrightarrow 1$ correspondence between $A(V)$ -modules and (maximal) V -modules generated by a space of least conformal weight.

Zhu's algebra and the classification of VOA modules

Def: Zhu's algebra

- Let $O(V)$ be the span of $\text{Res}(Y(a; z)b \frac{(1+z)^{h_a}}{z^2})$, $a \in V[h_a]$, $b \in V$.
- Zhu's algebra $A(V) = V/O(V)$ is a unital associative algebra with product induced by $a * b = \text{Res}(Y(a; z)b \frac{(1+z)^{h_a}}{z})$.

Thm: Zhu

- unit $[|0\rangle] = 1$, T in centre.
- Let M be a V -module with a least conformal weight h_0 . $\overline{M} = M[h_0]$, then $[a] \in A(V)$ acts on $\overline{M}$ as a_0 and $[a * b]$ as $a_0 b_0$.
- There exists a $1 \leftrightarrow 1$ correspondence between $A(V)$ -modules and (maximal) V -modules generated by a space of least conformal weight.
- If I is an ideal s.t. $|0\rangle, T \notin I$, then $A(I) = I/(O(V) \cap I)$ is an ideal in $A(V)$ and $A(V/I) = A(V)/A(I)$.

Example: Yang-Lee model

- central charge $c = -\frac{22}{5}$ Virasoro VOA

Example: Yang-Lee model

- central charge $c = -\frac{22}{5}$ Virasoro VOA
- maximal proper ideal generated by a singular vector χ
 $L_n \chi = 0, n \geq 1$

$$\chi = (L_{-2}^2 - \frac{3}{5}L_{-4})|0\rangle$$

Example: Yang-Lee model

- central charge $c = -\frac{22}{5}$ Virasoro VOA
- maximal proper ideal generated by a singular vector χ
 $L_n\chi = 0, n \geq 1$

$$\begin{aligned}\chi &= (L_{-2}^2 - \frac{3}{5}L_{-4})|0\rangle \\ \chi_0 &= L_0(L_0 + \frac{1}{5})\end{aligned}$$

Example: Yang-Lee model

- central charge $c = -\frac{22}{5}$ Virasoro VOA
- maximal proper ideal generated by a singular vector χ
 $L_n \chi = 0, n \geq 1$

$$\begin{aligned}\chi &= (L_{-2}^2 - \frac{3}{5}L_{-4})|0\rangle \\ \chi_0 &= L_0(L_0 + \frac{1}{5})\end{aligned}$$

- Least conformal weight can only be 0 or $\frac{-1}{5}$.

Example: Yang-Lee model

- central charge $c = -\frac{22}{5}$ Virasoro VOA
- maximal proper ideal generated by a singular vector χ
 $L_n\chi = 0, n \geq 1$

$$\begin{aligned}\chi &= (L_{-2}^2 - \frac{3}{5}L_{-4})|0\rangle \\ \chi_0 &= L_0(L_0 + \frac{1}{5})\end{aligned}$$

- Least conformal weight can only be 0 or $-\frac{1}{5}$.

Singular vectors are **very** hard to find in general. No general closed formulae are known.

Example: Yang-Lee model

- central charge $c = -\frac{22}{5}$ Virasoro VOA
- maximal proper ideal generated by a singular vector χ
 $L_n \chi = 0, n \geq 1$

$$\begin{aligned}\chi &= (L_{-2}^2 - \frac{3}{5}L_{-4})|0\rangle \\ \chi_0 &= L_0(L_0 + \frac{1}{5})\end{aligned}$$

- Least conformal weight can only be 0 or $\frac{-1}{5}$.

Singular vectors are **very** hard to find in general. No general closed formulae are known.

Goal for the rest of the talk: Find a better way to get these constraints for the example of the minimal models.

Outline

- 1 Chiral algebra basics
- 2 Free field theories and screening operators**
- 3 Symmetric polynomials
- 4 Application to the minimal models

The free boson

Def: Heisenberg algebra

$$\mathfrak{h} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C} a_n, [a_m, a_n] = m \delta_{m+n, 0}$$

The free boson

Def: Heisenberg algebra

$$\mathfrak{h} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C} a_n, [a_m, a_n] = m \delta_{m+n, 0}$$

Def: Heisenberg VOA

- $\mathcal{H} = \mathbb{C}[a_{-1}, a_{-2}, \dots] |0\rangle, a_n |0\rangle = 0, n \geq 0$
- $T = \frac{1}{2}(a_{-1}^2 + \alpha_0 a_{-2}) |0\rangle, c = 1 - 3\alpha_0^2$
- $Y(a_{-1} |0\rangle; z) = a(z) = \sum_n a_n z^{-n-1}, Y(T; z) = \frac{1}{2} : a(z)^2 : + \frac{\alpha_0}{2} \partial a(z)$

The free boson

Def: Heisenberg algebra

$$\mathfrak{h} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C} a_n, [a_m, a_n] = m \delta_{m+n, 0}$$

Def: Heisenberg VOA

- $\mathcal{H} = \mathbb{C}[a_{-1}, a_{-2}, \dots] |0\rangle, a_n |0\rangle = 0, n \geq 0$
- $T = \frac{1}{2}(a_{-1}^2 + \alpha_0 a_{-2}) |0\rangle, c = 1 - 3\alpha_0^2$
- $Y(a_{-1} |0\rangle; z) = a(z) = \sum_n a_n z^{-n-1}, Y(T; z) = \frac{1}{2} : a(z)^2 : + \frac{\alpha_0}{2} \partial a(z)$

For $(p_+, p_-) = 1, p_+, p_- \geq 2$, let $\alpha_0 = \alpha_+ + \alpha_-$,
 $\alpha_+ = \sqrt{\frac{2p_-}{p_+}}, \alpha_- = -\sqrt{\frac{2p_+}{p_-}}$, then $c = 1 - 6 \frac{(p_+ - p_-)^2}{p_+ p_-}$ is the central charge of the minimal models.

The free boson

Def: Heisenberg algebra

$$\mathfrak{h} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C} a_n, [a_m, a_n] = m \delta_{m+n, 0}$$

Def: Heisenberg VOA

- $\mathcal{H} = \mathbb{C}[a_{-1}, a_{-2}, \dots] |0\rangle, a_n |0\rangle = 0, n \geq 0$
- $T = \frac{1}{2}(a_{-1}^2 + \alpha_0 a_{-2}) |0\rangle, c = 1 - 3\alpha_0^2$
- $Y(a_{-1} |0\rangle; z) = a(z) = \sum_n a_n z^{-n-1}, Y(T; z) = \frac{1}{2} : a(z)^2 : + \frac{\alpha_0}{2} \partial a(z)$

For $(p_+, p_-) = 1, p_+, p_- \geq 2$, let $\alpha_0 = \alpha_+ + \alpha_-$,
 $\alpha_+ = \sqrt{\frac{2p_-}{p_+}}, \alpha_- = -\sqrt{\frac{2p_+}{p_-}}$, then $c = 1 - 6 \frac{(p_+ - p_-)^2}{p_+ p_-}$ is the central charge of the minimal models.

The minimal models can be realised as a subquotient of either the Virasoro VOA or the Heisenberg VOA.

The free boson

The simple $\mathcal{H}$ -modules are given by Fock modules.

Def: Fock modules $F_\lambda, \lambda \in \mathbb{C}$

- $F_\lambda = \mathbb{C}[a_{-1}, a_{-2}, \dots]|\lambda\rangle, a_0|\lambda\rangle = \lambda|\lambda\rangle, a_n|\lambda\rangle = 0, n \geq 1$
- $L_0|\lambda\rangle = \frac{1}{2}(a_0^2 - \alpha_0 a_0)|\lambda\rangle = h_\lambda|\lambda\rangle$
 $h_\lambda = \frac{1}{2}\lambda(\lambda - \alpha_0)$

The free boson

The simple $\mathcal{H}$ -modules are given by Fock modules.

Def: Fock modules $F_\lambda, \lambda \in \mathbb{C}$

- $F_\lambda = \mathbb{C}[a_{-1}, a_{-2}, \dots]|\lambda\rangle, a_0|\lambda\rangle = \lambda|\lambda\rangle, a_n|\lambda\rangle = 0, n \geq 1$
- $L_0|\lambda\rangle = \frac{1}{2}(a_0^2 - \alpha_0 a_0)|\lambda\rangle = h_\lambda|\lambda\rangle$
 $h_\lambda = \frac{1}{2}\lambda(\lambda - \alpha_0)$

Def: Generalised vertex operator

- new operator $\hat{a}: [a_n, \hat{a}] = \delta_{n,0}$

The free boson

The simple $\mathcal{H}$ -modules are given by Fock modules.

Def: Fock modules $F_\lambda, \lambda \in \mathbb{C}$

- $F_\lambda = \mathbb{C}[a_{-1}, a_{-2}, \dots]|\lambda\rangle, a_0|\lambda\rangle = \lambda|\lambda\rangle, a_n|\lambda\rangle = 0, n \geq 1$
- $L_0|\lambda\rangle = \frac{1}{2}(a_0^2 - \alpha_0 a_0)|\lambda\rangle = h_\lambda|\lambda\rangle$
 $h_\lambda = \frac{1}{2}\lambda(\lambda - \alpha_0)$

Def: Generalised vertex operator

- new operator $\hat{a}: [a_n, \hat{a}] = \delta_{n,0}$
- $e^{\lambda\hat{a}}|\mu\rangle = |\lambda + \mu\rangle$

The free boson

The simple $\mathcal{H}$ -modules are given by Fock modules.

Def: Fock modules $F_\lambda, \lambda \in \mathbb{C}$

- $F_\lambda = \mathbb{C}[a_{-1}, a_{-2}, \dots]|\lambda\rangle, a_0|\lambda\rangle = \lambda|\lambda\rangle, a_n|\lambda\rangle = 0, n \geq 1$
- $L_0|\lambda\rangle = \frac{1}{2}(a_0^2 - \alpha_0 a_0)|\lambda\rangle = h_\lambda|\lambda\rangle$
 $h_\lambda = \frac{1}{2}\lambda(\lambda - \alpha_0)$

Def: Generalised vertex operator

- new operator $\hat{a}$: $[a_n, \hat{a}] = \delta_{n,0}$
- $e^{\lambda\hat{a}}|\mu\rangle = |\lambda + \mu\rangle$
- $Y(|\lambda\rangle; z) = e^{\lambda\hat{a}}z^{\lambda a_0} \prod_{m \geq 1} e^{\lambda \frac{a_{-m}}{m} z^m} \prod_{m \geq 1} e^{-\lambda \frac{a_m}{m} z^{-m}}$

The free boson

The simple $\mathcal{H}$ -modules are given by Fock modules.

Def: Fock modules $F_\lambda, \lambda \in \mathbb{C}$

- $F_\lambda = \mathbb{C}[a_{-1}, a_{-2}, \dots]|\lambda\rangle, a_0|\lambda\rangle = \lambda|\lambda\rangle, a_n|\lambda\rangle = 0, n \geq 1$
- $L_0|\lambda\rangle = \frac{1}{2}(a_0^2 - \alpha_0 a_0)|\lambda\rangle = h_\lambda|\lambda\rangle$
 $h_\lambda = \frac{1}{2}\lambda(\lambda - \alpha_0)$

Def: Generalised vertex operator

- new operator $\hat{a}$: $[a_n, \hat{a}] = \delta_{n,0}$
- $e^{\lambda\hat{a}}|\mu\rangle = |\lambda + \mu\rangle$
- $Y(|\lambda\rangle; z) = e^{\lambda\hat{a}}z^{\lambda a_0} \prod_{m \geq 1} e^{\lambda \frac{a_{-m}}{m} z^m} \prod_{m \geq 1} e^{-\lambda \frac{a_m}{m} z^{-m}}$
- $T(w)Y(|\lambda\rangle; z) = \frac{h_\lambda}{(w-z)^2}Y(|\lambda\rangle; z) + \frac{1}{w-z}\partial Y(|\lambda\rangle; z) + \text{reg}$

The free boson

The simple $\mathcal{H}$ -modules are given by Fock modules.

Def: Fock modules $F_\lambda, \lambda \in \mathbb{C}$

- $F_\lambda = \mathbb{C}[a_{-1}, a_{-2}, \dots]|\lambda\rangle, a_0|\lambda\rangle = \lambda|\lambda\rangle, a_n|\lambda\rangle = 0, n \geq 1$
- $L_0|\lambda\rangle = \frac{1}{2}(a_0^2 - \alpha_0 a_0)|\lambda\rangle = h_\lambda|\lambda\rangle$
 $h_\lambda = \frac{1}{2}\lambda(\lambda - \alpha_0)$

Def: Generalised vertex operator

- new operator $\hat{a}$: $[a_n, \hat{a}] = \delta_{n,0}$
- $e^{\lambda\hat{a}}|\mu\rangle = |\lambda + \mu\rangle$
- $Y(|\lambda\rangle; z) = e^{\lambda\hat{a}}z^{\lambda a_0} \prod_{m \geq 1} e^{\lambda \frac{a_{-m}}{m} z^m} \prod_{m \geq 1} e^{-\lambda \frac{a_m}{m} z^{-m}}$
- $T(w)Y(|\lambda\rangle; z) = \frac{h_\lambda}{(w-z)^2}Y(|\lambda\rangle; z) + \frac{1}{w-z}\partial Y(|\lambda\rangle; z) + \text{reg}$
- $Y(|\lambda\rangle; z) : F_\mu \rightarrow F_{\lambda+\mu}[[z, z^{-1}]]z^{\lambda\mu}$

Def: Screening operator

- If $h_\lambda = 1$ (in this example $\lambda = \alpha_\pm$) then the field $S(z) = Y(|\lambda\rangle; z)$ is called a screening operator.

Def: Screening operator

- If $h_\lambda = 1$ (in this example $\lambda = \alpha_\pm$) then the field $S(z) = Y(|\lambda\rangle; z)$ is called a screening operator.
- $T(w)S(z) = \frac{1}{(w-z)^2}S(z) + \frac{1}{w-z}\partial S(z) + \text{reg} = \partial_z \frac{S(z)}{w-z} + \text{reg}$

Def: Screening operator

- If $h_\lambda = 1$ (in this example $\lambda = \alpha_\pm$) then the field $S(z) = Y(|\lambda\rangle; z)$ is called a screening operator.
- $T(w)S(z) = \frac{1}{(w-z)^2}S(z) + \frac{1}{w-z}\partial S(z) + \text{reg} = \partial_z \frac{S(z)}{w-z} + \text{reg}$
- Zero mode S_0 is a Virasoro homomorphism:
 $[T(w), S_0] = \text{Res}_{z=w} [T(w)S(z)] = 0$

Def: Screening operator

- If $h_\lambda = 1$ (in this example $\lambda = \alpha_\pm$) then the field $S(z) = Y(|\lambda\rangle; z)$ is called a screening operator.
- $T(w)S(z) = \frac{1}{(w-z)^2}S(z) + \frac{1}{w-z}\partial S(z) + \text{reg} = \partial_z \frac{S(z)}{w-z} + \text{reg}$
- Zero mode S_0 is a Virasoro homomorphism:
 $[T(w), S_0] = \text{Res}_{z=w} [T(w)S(z)] = 0$
- The map $S_0 : F_\mu \rightarrow F_{\mu+\lambda}$ is only well defined if $\lambda\mu \in \mathbb{Z}$.

Def: Screening operator

- If $h_\lambda = 1$ (in this example $\lambda = \alpha_\pm$) then the field $S(z) = Y(|\lambda\rangle; z)$ is called a screening operator.
- $T(w)S(z) = \frac{1}{(w-z)^2}S(z) + \frac{1}{w-z}\partial S(z) + \text{reg} = \partial_z \frac{S(z)}{w-z} + \text{reg}$
- Zero mode S_0 is a Virasoro homomorphism:
 $[T(w), S_0] = \text{Res}_{z=w} [T(w)S(z)] = 0$
- The map $S_0 : F_\mu \rightarrow F_{\mu+\lambda}$ is only well defined if $\lambda\mu \in \mathbb{Z}$.
- If $\lambda\mu \in \mathbb{Z}$ and $S_0|\mu\rangle \neq 0$ then $S_0|\mu\rangle$ is a **singular vector** in $F_{\mu+\lambda}$

Composing screening operators

If $\lambda\mu$ is not an integer, we must consider higher powers of the screening operator. Recall $\lambda = \alpha_{\pm}$, $\alpha_+ = \sqrt{\frac{2p_-}{p_+}}$, $\alpha_- = -\sqrt{\frac{2p_+}{p_-}}$.

$\lambda = \alpha_+$ **case:**

$$S_+(z_1) \cdots S_+(z_r) =$$

$$e^{r\alpha_+ \hat{a}} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{\alpha_+ a_0} \prod_{m \geq 1} e^{\alpha_+ \frac{a_{-m}}{m} \sum_i z_i^m} \prod_{m \geq 1} e^{-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m}}$$

Composing screening operators

If $\lambda\mu$ is not an integer, we must consider higher powers of the screening operator. Recall $\lambda = \alpha_{\pm}$, $\alpha_+ = \sqrt{\frac{2p_-}{p_+}}$, $\alpha_- = -\sqrt{\frac{2p_+}{p_-}}$.

$\lambda = \alpha_+$ case:

$$S_+(z_1) \cdots S_+(z_r) =$$

$$e^{r\alpha_+ \hat{a}} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{\alpha_+ a_0} \prod_{m \geq 1} e^{\alpha_+ \frac{a_{-m}}{m} \sum_i z_i^m} \prod_{m \geq 1} e^{-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m}}$$

Thm: Tsuchiya & Kanie, Varchenko

If and only if $\mu = \frac{1-r}{2}\alpha_+ + \frac{k}{\alpha_+}$, $k \in \mathbb{Z}$ then there exists a unique choice of a cycle Γ such that

$$S_+^{[r]} = \int_{\Gamma} S_+(z_1) \cdots S_+(z_r) dz_1 \cdots dz_r : F_{\frac{1-r}{2}\alpha_+ + \frac{k}{\alpha_+}} \rightarrow F_{\frac{1+r}{2}\alpha_+ + \frac{k}{\alpha_+}}$$

is a non-trivial Virasoro homomorphism.

- if $k \geq 0$ then $S_+^{[r]} | \frac{1-r}{2}\alpha_+ + \frac{k}{\alpha_+} \rangle = 0$.
- if $k < 0$ then $S_+^{[r]} | \frac{1-r}{2}\alpha_+ + \frac{k}{\alpha_+} \rangle \neq 0$.

The singular vector in the Heisenberg VOA

The singular vector that generates the ideal by which one must quotient the Heisenberg VOA to get the minimal model VOA is just

$$S_+^{[p_+-1]}|(1-p_+)\alpha_+\rangle \in \mathcal{H}$$

or

$$S_-^{[p_- -1]}|(1-p_-)\alpha_-\rangle \in \mathcal{H}.$$

The singular vector in the Heisenberg VOA

The singular vector that generates the ideal by which one must quotient the Heisenberg VOA to get the minimal model VOA is just

$$S_+^{[p_+-1]}|(1-p_+)\alpha_+\rangle \in \mathcal{H}$$

or

$$S_-^{[p_- -1]}|(1-p_-)\alpha_-\rangle \in \mathcal{H}.$$

The constraints on the minimal model conformal weights are obtained by evaluating the action of the zero mode $(S_+^{[p_+-1]}|(1-p_+)\alpha_+\rangle)_0$ or $(S_-^{[p_- -1]}|(1-p_-)\alpha_-\rangle)_0$.

Appearance of symmetric polynomials

Action of $S_+^{[r]}$ on $F_{\frac{1-r}{2}\alpha_+ + \frac{k}{\alpha_+}}$:

$$S_+^{[r]} = e^{r\alpha_+ \hat{a}} \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{(1-r)\frac{\alpha_+^2}{2}} \prod_{i=1}^r z_i^{k+1} \\ \times \prod_{m \geq 1} e^{\alpha_+ \frac{a-m}{m} \sum_i z_i^m} \prod_{m \geq 1} e^{-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m}} \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

Appearance of symmetric polynomials

Action of $S_+^{[r]}$ on $F_{\frac{1-r}{2}\alpha_+ + \frac{k}{\alpha_+}}$:

$$S_+^{[r]} = e^{r\alpha_+ \hat{a}} \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{(1-r)\frac{\alpha_+^2}{2}} \prod_{i=1}^r z_i^{k+1} \\ \times \prod_{m \geq 1} e^{\alpha_+ \frac{a-m}{m} \sum_i z_i^m} \prod_{m \geq 1} e^{-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m}} \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

This integral defines a well known inner product on the ring of symmetric polynomials $\mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$:

$$\langle f, g \rangle_r^{\alpha_+^2/2} = \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{(1-r)\frac{\alpha_+^2}{2}} f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

Appearance of symmetric polynomials

Action of $S_+^{[r]}$ on $F_{\frac{1-r}{2}\alpha_+ + \frac{k}{\alpha_+}}$:

$$S_+^{[r]} = e^{r\alpha_+ \hat{a}} \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{(1-r)\frac{\alpha_+^2}{2}} \prod_{i=1}^r z_i^{k+1} \\ \times \prod_{m \geq 1} e^{\alpha_+ \frac{a_m}{m} \sum_i z_i^m} \prod_{m \geq 1} e^{-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m}} \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

This integral defines a well known inner product on the ring of symmetric polynomials $\mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$:

$$\langle f, g \rangle_r^{\alpha_+^2/2} = \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{(1-r)\frac{\alpha_+^2}{2}} f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

This inner product is symmetric and it vanishes if $\deg(f) \neq \deg(g)$.

Outline

- 1 Chiral algebra basics
- 2 Free field theories and screening operators
- 3 Symmetric polynomials**
- 4 Application to the minimal models

Symmetric polynomials

- One can consider symmetric polynomials in either a finite or infinite number of variables:
 - ▶ $\Lambda_r = \mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$
 - ▶ $\Lambda = \varprojlim_r \Lambda_r$

Symmetric polynomials

- One can consider symmetric polynomials in either a finite or infinite number of variables:
 - ▶ $\Lambda_r = \mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$
 - ▶ $\Lambda = \varprojlim_r \Lambda_r$
- The power sums $p_k(z) = \sum_i z_i^k$ generate the ring of symmetric polynomials.
 - ▶ $\Lambda_r = \mathbb{C}[p_1, \dots, p_r]$
 - ▶ $\Lambda = \mathbb{C}[p_1, p_2, \dots]$

Symmetric polynomials

- One can consider symmetric polynomials in either a finite or infinite number of variables:
 - ▶ $\Lambda_r = \mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$
 - ▶ $\Lambda = \varprojlim_r \Lambda_r$
- The power sums $p_k(z) = \sum_i z_i^k$ generate the ring of symmetric polynomials.
 - ▶ $\Lambda_r = \mathbb{C}[p_1, \dots, p_r]$
 - ▶ $\Lambda = \mathbb{C}[p_1, p_2, \dots]$
- There is a well known basis that is orthogonal with respect to the inner product $\langle f, g \rangle_r^{\alpha_+^2/2}$ called Jack symmetric polynomials.

Jack symmetric polynomials

- Jack symmetric polynomials have 1 free parameter $\frac{\alpha_+^2}{2}$.

Jack symmetric polynomials

- Jack symmetric polynomials have 1 free parameter $\frac{\alpha_+^2}{2}$.
- For a fixed value of the parameter the individual polynomials are parametrised by partitions of integers. The degree of the polynomial is equal to the degree of the partition.

Jack symmetric polynomials

- Jack symmetric polynomials have 1 free parameter $\frac{\alpha+1}{2}$.
- For a fixed value of the parameter the individual polynomials are parametrised by partitions of integers. The degree of the polynomial is equal to the degree of the partition.
- If the length of the partition is greater than the number of variables, then the polynomial is 0.

Jack symmetric polynomials

- Jack symmetric polynomials have 1 free parameter $\frac{\alpha_+^2}{2}$.
- For a fixed value of the parameter the individual polynomials are parametrised by partitions of integers. The degree of the polynomial is equal to the degree of the partition.
- If the length of the partition is greater than the number of variables, then the polynomial is 0.
- The inner product of a Jack polynomial $P_\nu(z; \frac{\alpha_+^2}{2})$ with itself

$$\langle P_\nu(z; \frac{\alpha_+^2}{2}), P_\nu(z; \frac{\alpha_+^2}{2}) \rangle_r^{\alpha_+^2/2}$$

is a rational function in $\frac{\alpha_+^2}{2}$ and explicit formulae are known.

Jack symmetric polynomials

- Jack symmetric polynomials have 1 free parameter $\frac{\alpha_{\pm}^2}{2}$.
- For a fixed value of the parameter the individual polynomials are parametrised by partitions of integers. The degree of the polynomial is equal to the degree of the partition.
- If the length of the partition is greater than the number of variables, then the polynomial is 0.
- The inner product of a Jack polynomial $P_{\nu}(z; \frac{\alpha_{\pm}^2}{2})$ with itself

$$\langle P_{\nu}(z; \frac{\alpha_{\pm}^2}{2}), P_{\nu}(z; \frac{\alpha_{\pm}^2}{2}) \rangle_r^{\alpha_{\pm}^2/2}$$

is a rational function in $\frac{\alpha_{\pm}^2}{2}$ and explicit formulae are known.

- $$\prod_{m \geq 1} e^{\frac{\alpha_{\pm}^2}{2} \frac{p_m(y)p_m(z)}{m}} = \sum_{\nu} b_{\nu}(\frac{\alpha_{\pm}^2}{2}) P_{\nu}(y; \frac{\alpha_{\pm}^2}{2}) P_{\nu}(z; \frac{\alpha_{\pm}^2}{2})$$

Outline

- 1 Chiral algebra basics
- 2 Free field theories and screening operators
- 3 Symmetric polynomials
- 4 Application to the minimal models**

Explicit singular vector formulae (Mimachi & Yamada)

Let

$$\begin{aligned}\rho : \mathbb{C}[p_1, p_2, \dots] &\rightarrow \mathbb{C}[a_{-1}, a_{-2}, \dots] \\ p_k(y) &\mapsto \frac{2}{\alpha_+} a_{-k}.\end{aligned}$$

Then

$$S_+^{[p_+ - 1]} |(1 - p_+) \alpha_+ \rangle$$

Explicit singular vector formulae (Mimachi & Yamada)

Let

$$\begin{aligned}\rho : \mathbb{C}[p_1, p_2, \dots] &\rightarrow \mathbb{C}[a_{-1}, a_{-2}, \dots] \\ p_k(y) &\mapsto \frac{2}{\alpha_+} a_{-k}.\end{aligned}$$

Then

$$\begin{aligned}& S_+^{[p_+ - 1]} |(1 - p_+) \alpha_+ \rangle \\ &= \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^{p_+ - 1} z_i^{(2 - p_+) \frac{\alpha_+^2}{2}} \prod_{i=1}^{p_+ - 1} z_i^{1 - p_-} \prod_{m \geq 1} e^{\alpha_+ \frac{a_{-m}}{m} p_m(z)} |0\rangle \frac{dz_1 \cdots dz_{p_+ - 1}}{z_1 \cdots z_{p_+ - 1}}\end{aligned}$$

Explicit singular vector formulae (Mimachi & Yamada)

Let

$$\begin{aligned}\rho : \mathbb{C}[p_1, p_2, \dots] &\rightarrow \mathbb{C}[a_{-1}, a_{-2}, \dots] \\ p_k(y) &\mapsto \frac{2}{\alpha_+} a_{-k}.\end{aligned}$$

Then

$$\begin{aligned}& S_+^{[p_+-1]} |(1 - p_+) \alpha_+ \rangle \\&= \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^{p_+-1} z_i^{(2-p_+)\frac{\alpha_+^2}{2}} \prod_{i=1}^{p_+-1} z_i^{1-p_-} \prod_{m \geq 1} e^{\alpha_+ \frac{a_{-m}}{m} p_m(z)} |0\rangle \frac{dz_1 \cdots dz_{p_+-1}}{z_1 \cdots z_{p_+-1}} \\&= \langle \prod_{i=1}^{p_+-1} z_i^{p_--1}, \prod_{m \geq 1} e^{\alpha_+ \frac{a_{-m}}{m} p_m(z)} \rangle_{p_+-1}^{\alpha_+^2/2} |0\rangle\end{aligned}$$

Explicit singular vector formulae (Mimachi & Yamada)

Let

$$\begin{aligned}\rho : \mathbb{C}[p_1, p_2, \dots] &\rightarrow \mathbb{C}[a_{-1}, a_{-2}, \dots] \\ p_k(y) &\mapsto \frac{2}{\alpha_+} a_{-k}.\end{aligned}$$

Then

$$\begin{aligned}& S_+^{[p_+-1]} |(1 - p_+) \alpha_+ \rangle \\&= \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^{p_+-1} z_i^{(2-p_+)\frac{\alpha_+^2}{2}} \prod_{i=1}^{p_+-1} z_i^{1-p_-} \prod_{m \geq 1} e^{\alpha_+ \frac{a_{-m}}{m} p_m(z)} |0\rangle \frac{dz_1 \cdots dz_{p_+-1}}{z_1 \cdots z_{p_+-1}} \\&= \langle \prod_{i=1}^{p_+-1} z_i^{p_--1}, \prod_{m \geq 1} e^{\alpha_+ \frac{a_{-m}}{m} p_m(z)} \rangle_{p_+-1}^{\alpha_+^2/2} |0\rangle \\&= \sum_{\nu} \langle \prod_{i=1}^{p_+-1} z_i^{p_--1}, P_{\nu}(z; \frac{\alpha_+^2}{2}) \rangle_{p_+-1}^{\alpha_+^2/2} b_{\nu}(\frac{\alpha_+^2}{2}) \rho(P_{\nu}(y; \frac{\alpha_+^2}{2})) |0\rangle\end{aligned}$$

Explicit singular vector formulae (Mimachi & Yamada)

$$\begin{aligned}
 & S_+^{[p_+-1]} |(1-p_+)\alpha_+\rangle \\
 &= \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^{p_+-1} z_i^{(2-p_+)\frac{\alpha_+^2}{2}} \prod_{i=1}^{p_+-1} z_i^{1-p_-} \prod_{m \geq 1} e^{\alpha_+ \frac{a-m}{m} p_m(z)} |0\rangle \frac{dz_1 \cdots dz_{p_+-1}}{z_1 \cdots z_{p_+-1}} \\
 &= \left\langle \prod_{i=1}^{p_+-1} z_i^{p_--1}, \prod_{m \geq 1} e^{\alpha_+ \frac{a-m}{m} p_m(z)} \right\rangle_{p_+-1}^{\alpha_+^2/2} |0\rangle \\
 &= \sum_{\nu} \left\langle \prod_{i=1}^{p_+-1} z_i^{p_--1}, P_{\nu}(z; \frac{\alpha_+^2}{2}) \right\rangle_{p_+-1}^{\alpha_+^2/2} b_{\nu}(\frac{\alpha_+^2}{2}) \rho(P_{\nu}(y; \frac{\alpha_+^2}{2})) |0\rangle
 \end{aligned}$$

$$\prod_{i=1}^{p_+-1} z_i^{p_--1} = P_{(p_--1)^{p_+-1}}(z; \frac{\alpha_+^2}{2})$$

$$S_+^{[p_+-1]} |(1-p_+)\alpha_+\rangle = \rho(P_{(p_--1)^{p_+-1}}(y; \frac{\alpha_+^2}{2})) |0\rangle$$

Getting the conformal weights from the zero mode

The zero mode whose action we need to evaluate is the zero mode of the field

$$\begin{aligned} & S_+^{[p_+-1]} Y(|(1-p_+)\alpha_+\rangle; w) \\ &= \int_{\Gamma} S_+(z_1 + w) \cdots S_+(z_{p_+-1} + w) Y(|(1-p_+)\alpha_+\rangle; w) dz_1 \cdots dz_{p_+-1}. \end{aligned}$$

The easiest way to evaluate such a zero mode is to evaluate matrix elements

$$\langle \tau | S_+^{[p_+-1]} Y(|(1-p_+)\alpha_+\rangle; w) | \tau \rangle,$$

where $\tau \in \mathbb{C}$ and $\langle \tau | a_{-n} = 0, n \geq 1$.

Getting the conformal weights from the zero mode

$$\begin{aligned} \langle \tau | S_+^{[p_+-1]} Y(|(1-p_+)\alpha_+\rangle; w) | \tau \rangle &= \dots \\ &= \langle P_{(p_--1)p_+-1}(z; \frac{\alpha_+^2}{2}), \prod_{i=1}^{p_+-1} (1 + \frac{z_i}{w})^{\alpha_+ \tau} \rangle_{p_+-1}^{\alpha_+^2/2} \end{aligned}$$

Getting the conformal weights from the zero mode

$$\begin{aligned} \langle \tau | S_+^{[p_+-1]} Y(|(1-p_+)\alpha_+ \rangle; w) | \tau \rangle &= \dots \\ &= \langle P_{(p_--1)p_+-1}(z; \frac{\alpha_+^2}{2}), \prod_{i=1}^{p_+-1} (1 + \frac{z_i}{w})^{\alpha_+\tau} \rangle_{p_+-1}^{\alpha_+^2/2} \end{aligned}$$

There are known formulae for expanding

$$\prod_{i=1}^{p_+-1} (1 + \frac{z_i}{w})^{\alpha_+\tau} = \sum_{\nu} g_{\nu}(\alpha_+\tau) P_{\nu}(z; \frac{\alpha_+^2}{2}) w^{-|\nu|}$$

Getting the conformal weights from the zero mode

$$\begin{aligned} \langle \tau | S_+^{[p_+-1]} Y(|(1-p_+)\alpha_+ \rangle; w) | \tau \rangle &= \dots \\ &= \langle P_{(p_--1)p_+-1}(z; \frac{\alpha_+^2}{2}), \prod_{i=1}^{p_+-1} (1 + \frac{z_i}{w})^{\alpha_+\tau} \rangle_{p_+-1}^{\alpha_+^2/2} \end{aligned}$$

There are known formulae for expanding

$$\prod_{i=1}^{p_+-1} (1 + \frac{z_i}{w})^{\alpha_+\tau} = \sum_{\nu} g_{\nu}(\alpha_+\tau) P_{\nu}(z; \frac{\alpha_+^2}{2}) w^{-|\nu|}$$

$$\langle \tau | S_+^{[p_+-1]} Y(|(1-p_+)\alpha_+ \rangle; w) | \tau \rangle \propto \prod_{r,s} (h_{\tau} - h_{r,s})$$

$$h_{r,s} = h_{p_+-r, p_--s} = \frac{(p_+s - p_-r)^2 - (p_+ - p_-)^2}{4p_+p_-}$$

Applicability of these free field methods

- We saw today that symmetric polynomials can be used to “quite easily” classify the simple modules of the minimal models.
- Essentially the same method also works for free field realisations of affine $\mathfrak{sl}_2$.
- This method was a crucial tool in the classification of the simple modules of certain logarithmic extensions of the minimal models.

The End

Thank you for your attention!